Quiz Section 1

Review

1) Sum rule. If you can choose from EITHER one of n options, OR one of m options with NO overlap with the previous n, then the number of possible outcomes of the experiment is. _____.

2) Product rule. In a sequential process with m steps, if there are n_1 choices for the 1st step, n_2 choices for the 2nd step (given the first choice), ..., and n_m choices for the mth step (given the previous choices), then the total number of outcomes is _____.

3) Permutations. The number of ways to re-order n elements is _____.

4) k-permutations. The number of ways to choose a sequence of k distinct elements from a set of n elements is _____.

5) Set difference. Is it always true that $|A \backslash B| = |A| - |B|$?

Task 1 – Sets

a) For each one of the following sets, give its **cardinality**, i.e., indicate how many elements it contains:

- $A = \varnothing$ - $B = \{\varnothing\}$ - $C = \{\{\varnothing\}\}$ - $D = \{\varnothing, \{\varnothing\}\}$

b) Let $S = \{a, b, c\}$ and $T = \{c, d\}$. Compute:

- $S \cup T$ - $S \cap T$ - $S \backslash T$ - $2^{S \backslash T}$ - $S \times T$

Task 2 – Basic Counting

a) Credit-card numbers are made of 15 decimal digits, and a 16th checksum digit (which is uniquely determined by the first 15 digits). How many credit-card numbers are there?

b) How many positive divisors does $1440 = 2^5 3^2 5$ have?

c) How many ways are there to arrange the CSE 312 staff on a line (11 TAs, 2 professors) for a group picture?

d) How many ways are there to arrange the CSE 312 staff on a line so that Anup is at one of the ends or exactly in the middle?

Task 3 – Weird Card Game

In how many ways can a pack of fifty-two cards (in four suits of thirteen cards each) be dealt to thirteen players, four to each, so that every player has one card from each of the suits?

Task 4 – Seating

How many ways are there to seat 10 people, consisting of 5 couples, in a row of 10 seats if ...

a) ... all couples are to get adjacent seats?

b) ... anyone can sit anywhere, except that one couple insists on *not* sitting in adjacent seats?

Task 5 – HBCDEFGA

How many ways are there to permute the 8 letters A, B, C, D, E, F, G, H so that A is not at the beginning and H is not at the end?

Task 6 – Lizards and Snakes!

Loudon has three pet lizards, Rango, a gecko named Gordon, and a goanna named Joanna, as well as two small pet snakes, Kaa and Basilisk, but only 4 terrariums to put them in. In how many different ways can he put his 7 pets in these 4 terrariums so that no terrarium has both a snake and a lizard?

Task 7 – Birthday Cake

A chef is preparing desserts for the week, starting on a Sunday. On each day, only one of five desserts (apple pie, cherry pie, strawberry pie, pineapple pie, and cake) may be served. On Thursday there is a birthday, so cake must be served that day. On no two consecutive days can the chef serve the same dessert. How many dessert menus are there for the week?

Task 8 – Photographs

Suppose that 8 people, including you and a friend, line up for a picture. In how many ways can the photographer organize the line if she wants to have fewer than 2 people between you and your friend?

Task 9 – Extended Family Portrait

A group of n families, each with m members, are to be lined up for a photograph. In how many ways can the nm people be arranged if members of a family must stay together?